Palasani Sai Venkata Tanuj

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Skills

- **Languages**: C, C++, Python, Java
- **Web Technologies**: HTML, CSS, JavaScript
- **Tools/Platforms**: MySQL, Git, GitHub, Linux
- **Soft Skills**: Leadership, Communication, Critical Thinking, Time Management, Creativity

Projects

ML-Based Vehicle Load Monitoring System | GitHub May 2025

- Built a machine learning system to predict real-time vehicle load using Random Forest Regression with features like speed, axle count, suspension pressure, and vehicle type.
- Implemented anomaly detection using Isolation Forest to identify abnormal sensor behavior and overload conditions.
- Developed a Flask web app showing predicted load, overload alerts, anomaly notifications, and stored logs using SQLite.

Tech: Python, Flask, Scikit-learn, Random Forest, Isolation Forest, HTML, CSS, JavaScript, SQLite

AI ChoreMaster - Smart Family Task Manager | GitHub May 2025

- Developed an AI-powered task automation system supporting voice/text commands, smart task assignment, and completion tracking using Gemini AI.
- Implemented NLP-based date understanding, dual chatbot flow, speech recognition, speech synthesis, and analytics dashboard for family task patterns.
- Built a full web application with chore tracking, overdue alerts, family profiles, and persistent storage using SQLite.

Tech: HTML, CSS, JavaScript, Python, SQLite, Web Speech API, Speech Synthesis API, Gemini API

Intelligent CPU Scheduler Simulator | GitHub April 2025

- Developed a command-line CPU scheduling simulator demonstrating real OS scheduling behavior.
- Implemented FCFS, SJF (Preemptive & Non-Preemptive), Priority Scheduling, and Round Robin algorithms.
- Generated Gantt charts and calculated performance metrics like waiting time and turnaround time using a modular architecture.

Tech: C++, GCC, VS Code, GitHub

Training

Machine Learning with Data Science – Summer Training | Cipher Schools | *Certificate* June 2025 - July 2025

- Completed structured training covering Python, data manipulation (Pandas, NumPy), data visualization (Matplotlib, Seaborn), and machine learning algorithms including regression, classification, clustering, and anomaly detection.
- Worked extensively with tools like Google Colab, Git, Scikit-learn, and Excel for hands-on assignments and mini projects.
- Developed and implemented a full ML pipeline as part of the final project: data preprocessing, EDA, Random Forest Regression, Isolation Forest anomaly detection, and model evaluation.

Tech Covered: Python, Pandas, NumPy, Scikit-Learn, Matplotlib, Seaborn, Excel, Power BI

Certificates

- Ethical Hacking (Elite Silver Certificate) | NPTEL | *Certificate* November 2025
- Master Generative AI & Generative AI Tools (ChatGPT & More) | Udemy | *Certificate* August 2025
- Mastering in C: Basic to Beyond | CSE Pathshala | *Certificate* February 2024

Achievements

- **Led a Programming Teaching Initiative**: July 2025
  Mentored students by teaching essential programming concepts.
- **Finalist – Hack-a-Throne 1.0 Hackathon:** October 2024
  Reached the final round of the hackathon organized by GeeksforGeeks.

Education

- **Lovely Professional University** Punjab, India
  Bachelor of Technology - Computer Science and Engineering; ***CGPA: 8.71*** *Since August 2023*
- **Aditya Junior College** Amalapuram, Andhra Pradesh
  Intermediate; ***Percentage: 93%*** *June 2021 - April 2023*
- **Sri Chaitanya Techno School** Amalapuram, Andhra Pradesh
  Matriculation; ***Percentage: 99%*** *June 2020 - May 2021*